

- [REDACTED]

6.3 Head and Neck Cancer

◆ Key Features

- Head and neck cancer comprises a heterogeneous group of tumors, consisting predominantly of squamous cell carcinoma (SCC) of the upper aerodigestive tract.
- This type of cancer is linked to tobacco smoking and alcohol.
- There is a propensity for second primary tumors (between 4 and 7% per year), especially if the patient is still smoking.
- Multimodality treatment includes surgery, radiation, and chemotherapy.
- Adult with persistent neck mass is SCC until proved otherwise.

Head and neck cancer, predominantly SCC, can affect the oral cavity, pharynx, larynx, hypopharynx, cervical esophagus, nose, and paranasal sinuses. The goal of treatment is cure or palliation with preservation of function. Specific sites and subsites of head and neck cancer are discussed in subsequent chapters.

Table 6.5 Tumor differentiation grading using the Broder classification (tumor grade [G])

G1: Well differentiated
G2: Moderately well differentiated
G3: Poorly differentiated
G4: Undifferentiated

Note: There is no statistically significant correlation between degree of differentiation and the biologic behavior of the cancer; however, vascular invasion is a negative prognostic factor.

◆ Epidemiology

Head and neck SCC accounts for ~ 5% of cancers in the United States. This corresponds to an estimated 17 per 100,000 Americans per year with newly diagnosed SCC of the head and neck. These cancers are more common in men and typically occur in patients over age 50. The etiology includes tobacco use (smoking and smokeless) and alcohol consumption. Eighty-five percent of head and neck SCC are linked to tobacco use. The synergistic effect of alcohol and smoking increases the risk of disease many more times than the simple

additive risk of either risk factor alone. In SCC, mutations in the *TP53* gene correlate with drinking and smoking habits. Approximately 15% of patients have a viral etiology. Epstein-Barr virus (EBV) has been implicated in the development of nasopharyngeal carcinoma. Human papillomavirus (HPV) infection is another factor implicated in the carcinogenesis of upper aerodigestive tract tumors. In particular, HPV-16 can be isolated in up to 72% of oropharyngeal cancers. The recent increase in cancers of the tongue and tonsils in developed countries, particularly in younger patients, has been associated with HPV.

During the past 20 years, the overall incidence of head and neck SCC has been declining in the United States, a decline that is attributed to a decrease in the prevalence of smoking. In other parts of the world, head and neck SCC is attributed to habitual and cultural habits such as chewing *paan* (betel leaf with areca nut), smoking *khat* (*Catha edulis*), and drinking *yerba maté*.

◆ Clinical

Signs

Signs may include hoarseness, muffled speech, trismus, and recurrent epistaxis. Many patients present with a neck mass as chief complaint, representing metastatic nodal disease from an occult primary tumor in the upper aerodigestive tract (**Table 6.6**).

Table 6.6 Incidence of cervical lymph node metastases associated with carcinomas of the upper aerodigestive tract

Tumor location	Incidence of cervical lymph node metastases at diagnosis (%)
Oral cavity	30–65
Oropharynx	39–83
Nasopharynx	60–90
Hypopharynx	52–72
Supraglottis	35–54
Glottis	7–9
Nasal cavity and paranasal sinuses	10–20
Salivary glands	25–50
Thyroid gland	18–84*

*Depends on age and histologic subtype. (Adapted with permission from Probst R, Grevers G, Iro H. *Basic Otorhinolaryngology: A Step-by-Step Learning Guide*. Stuttgart/New York: Thieme; 2006:333.)

Symptoms

Symptoms of head and neck SCC are variable and depend on the site and stage of the primary tumor (see “Staging of Head and Neck Cancer”). Early symptoms may be vague and mimic benign disease and are therefore

discovered only at advanced stages of disease. Symptoms may include dysphagia, odynophagia, a globus sensation, changes in voice (this includes both hoarseness and velopharyngeal insufficiency), referred otalgia, CN hypoesthesia, nasal obstruction, epiphora, and hyposmia.

Differential Diagnosis

- Upper respiratory infections such as pharyngitis, laryngitis, deep neck infections or abscesses
- Congenital masses and cysts
- Upper airway manifestations of rheumatologic and autoimmune diseases
- Hematologic malignancies (lymphoma)
- Tuberculosis
- Fungal infections

◆ Evaluation

History

History should include questions about risk factors, breathy voice or prolonged hoarseness, dysphagia, hemoptysis, difficulty breathing, shortness of breath, otalgia, and unintentional weight loss, fevers, or chills ("B symptoms").

Physical Exam

Physical exam should include careful inspection of the oral and oropharyngeal mucosa for lesions and palpation of the tonsillar region and tongue base for firm nodules or masses.

An indirect mirror or flexible fiberoptic laryngoscopy should be performed. During this examination, the patient should be asked to perform several maneuvers such as tongue protrusion, puffing out the cheeks, lightly coughing, and speaking to better visualize and access the larynx and the hypopharynx. Specifically, these assessments are done to assess for asymmetry or the presence of any visible masses or ulcerations. It is also important that laryngeal motility be assessed, as this is critical in tumor staging. The neck should be examined in a systematic fashion. Any palpable lymph nodes should be assessed with regard to size, location, texture, and mobility.

Imaging

A contrast-enhanced computed tomography (CT) or magnetic resonance imaging (MRI) scan of head and neck should be obtained to assess local and regional extent of disease and involvement of adjacent structures, such as the great vessels of the neck and the prevertebral fascia. Specific characteristics of regional lymphadenopathy, if present, should be noted, such as extracapsular spread, central necrosis, and size of involved lymph nodes. Criteria for pathologic lymph nodes include nodes that have a round shape with loss of a hilum, presence of central necrosis, and/or lymph nodes larger than 10 mm (15 mm in level 1).

A metastatic work-up may consist of a chest radiograph with liver function tests, CT scan of the chest and abdomen, or, alternatively, a fluorodeoxyglucose–positron emission tomography (FDG–PET) scan.

Labs

Blood count, electrolyte, and liver function tests should be performed to assess nutritional status.

Other Tests

Fine-needle aspiration biopsy (FNAB) is a highly accurate technique for the investigation of cervical lymph node metastases in head and neck SCC and is the first-line test in a patient with lymphadenopathy. Sensitivity of this test is improved when performed with ultrasound guidance. It is best to avoid open biopsy of a neck mass, because of the problems of tumor spillage and violation of fascial planes.

Patients in whom a suspicion of head and neck SCC exists should undergo biopsy of suspicious primary site lesions. Because of the propensity for second primary tumors that accompany head and neck SCC, these patients should undergo a panendoscopy (triple endoscopy), consisting of laryngoscopy, esophagoscopy, and bronchoscopy together, to search for simultaneous lesions. This may also lend insight into the extent of the primary lesion, particularly important in the smoking patient. This is a point of controversy, and PET scanning may play a greater role in the future. Mention should be made that in the current literature, the sensitivity of PET scanning is not adequate to replace panendoscopy: for instance, the larynx has baseline activity on PET due to patient breathing and/or talking during the scan.

Pathology

Ninety percent of head and neck cancers are SCCs.

◆ Treatment Options

Treatment of head and neck SCC consists of surgery, radiotherapy, chemotherapy, or a combination of these. Surgical resection remains the gold standard for treatment of head and neck cancer. Surgery may address the primary tumor as well as cervical metastasis.

Radiation therapy for SCC of the head and neck involves the delivery of high-energy ionizing radiation to targeted tissues. Radiation doses can be delivered by different methods, including fractionation, hyperfractionation, accelerated fractionation, and intensity-modulated radiotherapy (IMRT).

Chemotherapy for SCC of the head and neck involves the systemic administration of cytotoxic drugs that target rapidly dividing cells. Individual chemotherapeutic agents effective in the therapy of head and neck cancer include cisplatin, methotrexate, 5-fluorouracil, taxanes, ifosfamide, and bleomycin. Currently, chemotherapy is not a primary therapy for the treatment of head and neck cancer, but either it can be used as an induction agent followed by surgery and/or radiation, or it can be given concurrently with radiation treatment.

Of the molecularly targeted agents, cetuximab is an IgG1 chimeric antibody directed against the epidermal growth factor receptor (EGFR).

Other therapy methods for head and neck cancer include photosensitizers and interstitial laser therapy, photodynamic therapy, immune therapy, gene therapy, and targeted therapy against HPV.

◆ Cancer of Unknown Primary

Patients with head and neck cancer can initially present with a painless neck mass. In 2 to 8% of these patients, the tumor origin is not known. This is classified as a cancer of unknown primary (CUP), carcinoma of unknown primary origin, or occult primary malignancy. A primary tumor is considered unknown only *after* a thorough investigation (including physical exam, imaging, and biopsies) has been completed.

Treatment of CUP is controversial. Surgical excision in the form of a neck dissection followed by radiation therapy allows a lower total dose of radiation to be used. Primary radiation therapy provides treatment to both the upper aerodigestive tract and its locoregional metastasis but forces the radiation oncologist to treat a wider field, as the primary site is unknown. Chapter 6.3.8 discusses the management of this diagnosis in more detail.

◆ Outcome and Follow-Up

The treatment of head and neck SCC, whether surgical or chemoradiation, often leaves the patient with significant speech and swallowing deficits. Therefore, is it paramount for a multidisciplinary team to treat head and neck SCC. This team can include the surgical oncologist, medical and radiation oncologist, oral surgeon, prosthodontist, speech language and swallowing pathologist, nurse, and social worker.

Patients with regional neck disease prior to treatment should have a CT scan or an integrated FDG-PET/CT. There is controversy about the need for a planned neck dissection following radiotherapy in patients with high-risk disease. Most recurrences of SCC occur within 3 years of the initial treatment. The National Comprehensive Cancer Network (NCCN) guidelines suggest routine follow up after treatment. This includes follow up every 1 to 2 months the first year, every 2 to 6 months the second year, every 4 to 8 months the third through fifth years and annual follow up afterwards.

Patients should be told of the risk of a second primary tumor and encouraged to report any new symptoms. The risk of a second primary carcinoma is highest in those who continue to smoke. Patients should be counseled for tobacco cessation.

◆ Staging of Head and Neck Cancer

- With the publication and the 2018 implementation of the American Joint Committee on Cancer (AJCC) 8th edition, the classification of the regional lymph node metastases and their effect on stage grouping are fairly consistent throughout all anatomic sites of head and neck cancer (except nasopharynx and HPV-associated oropharynx cancer). Additionally, extranodal extension (ENE) has been included in the staging criteria. This can be determined clinically (fixation of the node to the skin or muscles, or nerve dysfunction) or pathologically and can be supported (but not determined) radiologically. HPV associated oropharyngeal cancers have separate nodal classification. For non-HPV associated cancers, the clinical and pathologic nodal (N) categories are as follows:

Regional Lymph Nodes (N)*

NX: Regional lymph nodes cannot be assessed

N0: No regional lymph node metastasis

N1: Metastasis to a single ipsilateral lymph node measuring ≤ 3 cm in greatest diameter and ENE negative

N2: Further divided into three categories:

N2a: Single ipsilateral lymph node > 3 and ≤ 6 cm and ENE negative (pathologic staging also includes a single ipsilateral or contralateral lymph node ≤ 3 cm and ENE positive)

N2b: Multiple ipsilateral lymph nodes ≤ 6 cm and ENE negative

N2c: Bilateral or contralateral lymph nodes ≤ 6 cm in greatest dimension and ENE negative

N3: Now further divided into two categories:

N3a: Metastasis to a lymph node > 6 cm, ENE negative

N3b: Metastasis to a single lymph node that is ENE positive (pathologic staging includes single lymph nodes that are ENE positive > 3 cm), or metastasis to multiple lymph nodes with any ENE positive

A designation of "U" or "L" can be attached to the N category to indicate nodes above ("U") or below ("L") the lower border of the cricoid. Similarly, clinical and pathologic ENE should be recorded as ENE (–) or ENE (+). Clinical and pathologic staging are similar with the differences noted in parenthesis.

*Superior mediastinal lymph nodes are considered regional lymph nodes (level VII). Midline nodes are considered ipsilateral nodes.

- Distant metastatic disease is divided into two categories:

M0: Absence of distant disease

M1: Presence of distant metastatic disease

- The T category of a tumor indicates the extent of the primary tumor and varies by anatomic subsite. This can be measured by size, as in the oral cavity, oropharynx, and salivary glands; by involvement of varying subsites, as in the nasopharynx, hypopharynx, and larynx; or by extent of invasion and destruction, as in the maxillary sinus. With the new edition, it is also effected by HPV status in the oropharynx. The specific T stage criteria are discussed in each subsite section.

- Across all anatomic sites of the head and neck (except the nasopharynx), the following classifications apply for HPV negative (p16 negative) cancers (**Table 6.7**):

Stage I disease: Includes only T1 N0 M0 tumors

Stage II disease: Includes only T2 N0 M0 tumors

Stage III disease: Includes T3 N0 M0 and T1–3 disease that is N1 M0

Stage IV disease: Includes T4 tumors with or without nodal disease, as well as any tumor with N2 or N3 disease or evidence of distant metastatic disease

Table 6.7 Cancer stage groupings

	N0	N1	N2	N3
T1	Stage I	Stage III	Stage IVA	Stage IVB
T2	Stage II	Stage III	Stage IVA	Stage IVB
T3	Stage III	Stage III	Stage IVA	Stage IVB
T4a	Stage IVA	Stage IVA	Stage IVA	Stage IVB
T4b	Stage IVB	Stage IVB	Stage IVB	Stage IVB
M1	Stage IVC	Stage IVC	Stage IVC	Stage IVC

Please refer to the site-specific sections for TNM categories and staging for nasopharynx and HPV-associated oropharynx cancers.

Table 6.8 Tumor differentiation grading using the Broder classification (tumor grade [G])

G1: Well differentiated
G2: Moderately well differentiated
G3: Poorly differentiated
G4: Undifferentiated

Note: There is no statistically significant correlation between degree of differentiation and the biologic behavior of the cancer; however, vascular invasion is a negative prognostic factor.

6.3.3 Sinonasal Cancer

◆ Key Features

- Sinonasal cancer initially may mimic benign sinus disease.
- Tumors of the paranasal sinuses often present with advanced disease.
- Cure rates are generally $\leq 50\%$.
- Most patients die of direct extension into vital areas.

Malignant tumors of the sinonasal tract are extremely rare, accounting for 0.2% of all invasive cancers and 3% of head and neck cancers. Cancers of the maxillary sinus are the most common. Tumors of the ethmoid sinuses are less common (20%), and cancers of the sphenoid and frontal sinuses are rare ($< 1\%$). Local extension often makes it difficult to access the sinus of origin.

◆ Epidemiology

Chemical carcinogens such as chromium, nickel, thorium dioxide, and tanning chemicals have been implicated in the development of carcinoma of the paranasal sinuses. Exposure to wood dust has been implicated specifically in adenocarcinoma of the ethmoid. Interestingly, tobacco use was previously thought not to play a role in sinonasal carcinogenesis. However, up to a fivefold increased risk of sinonasal carcinoma has been observed with heavy smoking. Rarely, sinonasal cancers may present as a second primary tumor in tobacco users with other head and neck cancers.

◆ Clinical

Signs and Symptoms

The clinical presentation of sinus malignancies is nonspecific and often mimics benign disease; thus, diagnosis is often delayed for months. Key indicators of malignancy are cranial neuropathies, proptosis, and pain of maxillary dentition; trismus; palatal and alveolar ridge fullness; or frank erosion into the oral cavity. Symptoms include nasal obstruction, discharge, stuffiness, congestion, recurrent epistaxis, unilateral tearing, diplopia, exophthalmos, infraorbital nerve hypoesthesia, cheek swelling, facial asymmetry, hearing loss, and serous otitis media due to nasopharyngeal extension.

Differential Diagnosis

The differential diagnosis includes benign sinus disease, benign sinus tumors, and metastatic tumors to the sinus.

◆ Evaluation

History

The patient history should include known carcinogen exposure, tobacco usage, and prolonged benign sinus symptoms and signs.

Physical Exam

A complete head and neck examination, including nasal endoscopy, should be performed. The sinonasal, ocular, and neurologic systems should be studied in detail. Evidence of nerve hypesthesia, diplopia, proptosis, and loose dentition should be carefully evaluated. Suspicious lesions should be biopsied.

Imaging

Imaging should include either a contrast-enhanced computed tomography (CT) scan or magnetic resonance imaging (MRI). There may be a role for integrated ^{18}F fludeoxyglucose (FDG) positron emission tomography (PET)/CT.

Other Tests

A definitive diagnosis requires a biopsy. Special attention should be paid to cranial nerve (CN) function because malignant paranasal tumors are associated with a high incidence of cranial neuropathies compared with inflammatory or benign sinus disease. Be cautious about in-office biopsy, as these tumors can have high vascularity.

Pathology

Squamous cell carcinoma (SCC) is the most frequent type of malignant tumor in the paranasal sinuses (70–80%). Minor salivary gland tumors constitute 10 to 15% of these neoplasms. Some 5% of cases are lymphomas. Other tumors include sinonasal undifferentiated carcinoma (SNUC), chondrosarcoma, osteosarcoma and malignant melanoma, and olfactory neuroblastoma (**Fig. 6.8**).

Inverted papilloma, a benign tumor with a tendency to recur (see **Chapter 4.4**), may transform into a malignant SCC of the paranasal sinuses in a small percentage of cases.

◆ Treatment Options

Most stage T1 or T2 maxillary sinus carcinomas are treated by surgery alone, provided adequate resection margins are obtained. This may be en bloc surgical resection or endoscopic sinus surgery, depending on the extent of disease and experience of the surgeon. The specific approach is determined by the location of disease and histology (**Fig. 6.9**).

T3 and T4 lesions are treated by combination therapy with surgery and radiation. The issue of whether radiation is more effective before or after surgery remains controversial. Chemotherapy alone is generally used as a palliative measure.

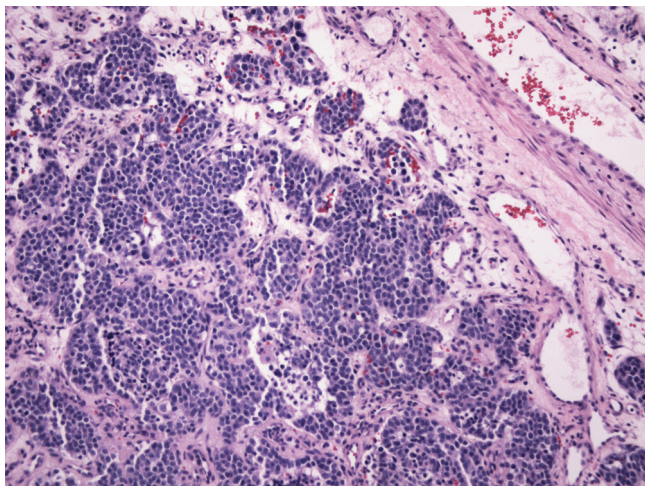


Fig. 6.8 Low-grade forms of olfactory neuroblastoma tumor have a distinct lobular array of small blue cells set in a neurofibrillary or vascular stroma. As the grade increases, the tumor becomes more solid, with increased nuclear pleomorphism and mitotic activity. (Used with permission from Har-El G, Day T, Nathan C-A, Nguyen SA. *A Multidisciplinary Approach to Head and Neck Neoplasms*. New York: Thieme; 2013. *Otorhinolaryngology—Head and Neck Surgery Series*.)

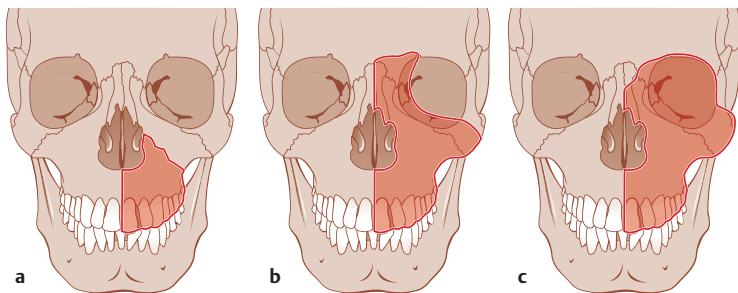


Fig. 6.9 Resection of the upper jaw. **(a)** Partial resection. **(b)** Total resection. **(c)** Total resection with exenteration of the orbit. (Used with permission from Behrbohm H, Kaschke O, Nawka T, Swift A. *Ear, Nose, and Throat Diseases: With Head and Neck Surgery*. 3rd ed. Stuttgart/New York: Thieme; 2009:227.)

◆ Outcome and Follow-Up

For maxillary tumors, malignancy behind Öhngren's plane is regarded to carry a much poorer prognosis because of the rapid spread to the orbit and middle cranial fossa (**Fig. 6.10**). Despite improvements in surgical ablative and reconstructive techniques, radiation delivery modalities, and imaging technologies, disease-free survival at 5 years remains < 50%, independent of stage. Five-year disease-free survival for patients with advanced-stage cancer drops to 25%.

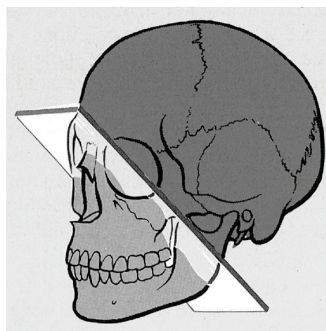


Fig. 6.10 Öhngren's plane passing through the medial canthus and the mandibular angle. It divides the maxillary sinus into a superoposterior part and an inferoanterior part. Cancer limited to the latter part typically carries a better prognosis. (Used with permission from Becker W, Naumann HH, Pfaltz CR. *Ear, Nose, and Throat Diseases: A Pocket Reference*. 2nd ed. Stuttgart/New York: Thieme; 1994:293.)

♦ Staging of Nose and Paranasal Sinus Cancer: For All Carcinomas Excluding Mucosal Malignant Melanoma

Primary Tumor (T): Maxillary Sinus

- TX: Cannot be assessed
- Tis: Carcinoma in situ
- T1: Tumor limited to the maxillary sinus mucosa with no erosion or destruction of bone
- T2: Tumor causing bone erosion or destruction, including extension into the hard palate and/or middle nasal meatus, except extension to posterior wall of maxillary sinus and pterygoid plates
- T3: Tumor invades any of the following: bone of the posterior wall of maxillary sinus, subcutaneous tissues, floor or medial wall of orbit, pterygoid fossa, or ethmoid sinuses
- T4a: Tumor invades anterior orbital contents, skin of cheek, pterygoid plates, infratemporal fossa, cribriform plate, or sphenoid or frontal sinuses
- T4b: Tumor invades any of the following: orbital apex, dura, brain, middle cranial fossa, cranial nerves other than maxillary division of trigeminal nerve (V_2), nasopharynx, or clivus

Primary Tumor (T): Nasal Cavity and Ethmoid Sinus

- TX: Cannot be assessed
- Tis: Carcinoma in situ
- T1: Tumor restricted to any one subsite, with or without bone invasion
- T2: Tumor invading two subsites in a single region or extending to involve an adjacent region within the nasoethmoidal complex, with or without bone invasion

- T3: Tumor extends to invade the medial wall or floor of the orbit, maxillary sinus, palate, or cribriform plate
- T4a: Tumor invades any of the following: anterior orbital contents, skin of nose or cheek, minimal extension to anterior cranial fossa, pterygoid plates, or sphenoid or frontal sinuses
- T4b: Tumor invades any of the following: orbital apex, dura, brain, middle cranial fossa, cranial nerves other than maxillary division of trigeminal nerve (V_2), nasopharynx, or clivus

Regional Lymph Nodes (N)*

- NX: Regional lymph nodes cannot be assessed
- N0: No regional lymph node metastasis
- N1: Metastasis to a single ipsilateral lymph node measuring ≤ 3 cm in greatest diameter and ENE negative
- N2: Further divided into three categories:
- N2a: Single ipsilateral lymph node > 3 and ≤ 6 cm and ENE negative (pathologic staging also includes a single ipsilateral or contralateral lymph node ≤ 3 cm and ENE positive)
 - N2b: Multiple ipsilateral lymph nodes ≤ 6 cm and ENE negative
 - N2c: Bilateral or contralateral lymph nodes ≤ 6 cm in greatest dimension and ENE negative
- N3: Now further divided into two categories:
- N3a: Metastasis to a lymph node > 6 cm, ENE negative
 - N3b: Metastasis to a single lymph node that is ENE positive (pathologic staging includes single lymph nodes that are ENE positive > 3 cm), or metastasis to multiple lymph nodes with any ENE positive

A designation of "U" or "L" can be attached to the N category to indicate nodes above ("U") or below ("L") the lower border of the cricoid. Similarly, clinical and pathologic ENE should be recorded as ENE (-) or ENE (+). Clinical and pathologic staging are similar with the differences noted in parenthesis.

*Superior mediastinal lymph nodes are considered regional lymph nodes (level VII). Midline nodes are considered ipsilateral nodes.

Distant Metastasis (M)

- M0: No distant metastasis
- M1: Distant metastasis

American Joint Committee on Cancer Stage Groupings of Sinonasal Cancers Except Mucosal Malignant Melanoma (8th edition)

- Stage 0 disease: Carcinoma in situ (TisN0M0)
- Stage I disease: Includes only T1 N0 M0 tumors
- Stage II disease: Includes only T2 N0 M0 tumors
- Stage III disease: Includes T3 N0 M0 and T1 -3 disease that is N1 M0

Stage IVA disease: Includes T4a disease with N0-N2 M0 disease and T1-T3 disease that is N2 M0

Stage IVB disease: Includes T4b disease with any nodal disease or T1-T4a disease with N3 disease without distant metastases (M0)

Stage IVC disease: Any disease with distant metastases (M1)

See **Table 6.7** in Chapter 6.3.3.

6.3.4 Nasopharyngeal Cancer

◆ Key Features

- There is a high frequency of nasopharyngeal cancer (NPC) among patients of Chinese ethnicity and descent.
- It is associated with Epstein-Barr virus (EBV) exposure.
- The diagnosis must be excluded in patients with asymptomatic cervical lymphadenopathy and unilateral serous otitis media.

Nasopharyngeal cancer (NPC) is a distinct type of head and neck cancer that differs from other malignancies of the upper aerodigestive tract with respect to epidemiology, pathology, clinical presentation, and responses to treatment. NPC is an uncommon neoplasm in most parts of the world but is endemic in East Asia. Seventy percent of patients with newly diagnosed NPC present with locally advanced disease. Latent EBV infection seems to be crucial in the pathogenesis of NPC. Studies have established that NPC cells express two distinct EBV latent membrane proteins: LMP-1 and LMP-2. These proteins are attractive targets for adaptive immunotherapy.

◆ Anatomy

The nasopharynx is bounded superiorly by the basi-occiput and basi-sphenoid, posteriorly by the C1 and C2 cervical bodies, anteriorly by the choanae, and inferiorly by the soft palate. The lateral walls are occupied primarily by the eustachian tube orifice. Immediately posterior to the eustachian tube orifice is the Rosenmüller fossa (pharyngeal recess), where most nasopharyngeal carcinomas originate.

◆ Epidemiology

NPC occurs most often in China, where it is the third most common malignancy among men, with an incidence rate of 15 to 50 per 100,000.

NPC rates are also high in Vietnamese and Filipino men. There is an intermediate incidence in Inuit (Eskimos) and in the populations of the Mediterranean basin. Emigration from high- to low-incidence areas reduces the incidence of NPC in first-generation Chinese, but incidence still remains

at seven times the rate seen in Caucasians. NPC occurs in younger patients, including children, but the peak incidence is seen in people aged 55 to 64 years. Males are affected three times more often than females.

◆ Clinical

Signs and Symptoms

Early signs and symptoms are subtle and variable and are often initially ignored by both patient and physician. Cervical lymphadenopathy is the most common sign (50–90% of patients), followed by blood-stained nasal discharge or epistaxis, unilateral serous otitis media, and cranial neuropathies (most often CN VI, followed by CN V). Five to seven percent of all patients have systemic metastases at presentation, most often to bone.

Symptoms include unilateral nasal obstruction, unilateral hearing loss and otalgia, diplopia, facial or neck pain, and paresthesia.

Differential Diagnosis

- Minor salivary gland tumors
- Juvenile nasopharyngeal angiofibroma
- Adenoid hypertrophy
- Tornwaldt cysts
- Fibromyxomatous polyps
- Choanal polyps, fibromas
- Papillomas
- Osseous/fibroosseous tumors
- Craniopharyngiomas
- Extracranial meningiomas
- Chordomas

◆ Evaluation

History

History should include questions about epistaxis, nasal obstruction and discharge, hearing loss or clogged ear, headache, diplopia, facial pain, and numbness.

Physical Exam

The physical exam should include a fiberoptic nasopharyngoscopy. The neck should be examined in a systematic fashion. Any lymph nodes should be assessed with regard to size, location, and mobility.

Imaging

Imaging is required for staging and treatment planning for NPC. Computed tomography (CT) and magnetic resonance imaging (MRI) are recommended for the diagnostic process and evaluation of tumoral extent and bone erosion and to delineate tumor extension into the parapharyngeal and retropharyngeal spaces, the oropharynx, the orbit, and the intracranial compartment.

A chest X-ray, liver ultrasound, and a bone scan are recommended for all patients with nodal disease. The role of positron emission tomography (PET) in the staging of nasopharyngeal carcinoma has not been well established.

Labs

The quantitative analysis of cell-free EBV DNA in plasma of patients has been studied, and data suggest a possible value of this tool in screening and monitoring treatment. High IgA antibody levels to EBV capsid antigen and early antigen (EA) provide a valuable screening tool for early cases in high-incidence populations.

Other Tests

A dental examination is required before instituting radiotherapy to reduce the development of postradiotherapy complications. Patients should continue with meticulous dental care and fluoride prophylaxis.

Pathology

Classic nasopharyngeal carcinoma has been classified into three types by the World Health Organization (WHO):

Type 1: Keratinizing squamous cell carcinoma (SCC)

Type 2 (2a): Nonkeratinizing or poorly differentiated carcinoma

Type 3 (2b): Undifferentiated carcinoma (lymphoepithelioma)

WHO types II and III exhibit between 82 and 100% positivity with respect to EBV antibody titers. Type 1 may have an association with cigarette and alcohol consumption and accounts for up to 30% of cases in nonendemic areas and < 5% in endemic areas. Several genetic markers of the human leukocyte antigen (HLA) system have been investigated in patients with NPC in China and other parts of Asia. HLA-A2 and HLA-B*57:01 were associated with an increased incidence, whereas HLA-A11 was associated with a decreased risk. See the histology slides in **Fig. 6.11**.

◆ Treatment Options

Medical

Radiotherapy is the cornerstone of the definitive treatment for NPC. This is because NPCs are particularly radiosensitive, whereas the tumor is in a relatively inaccessible location, making surgical excision difficult and highly morbid. External beam is most commonly delivered by opposed lateral fields to encompass the primary tumor and upper neck. Radiation doses of 70 to 76 Gy in fractions of 1.8 to 2.0 Gy per day to the primary and anatomic structures at risk within the vicinity of the nasopharynx. Because there is a high incidence of subclinical neck disease, radiation doses between 50 and 60 Gy are used to electively treat the neck.

Recent data shows a clear role for concomitant chemoradiotherapy followed by adjuvant chemotherapy, which provides statistically significant improvement in overall survival and disease-free survival.

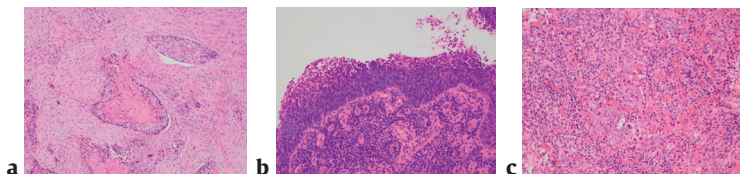


Fig. 6.11 (a) Histology slide showing type I squamous cell nasopharyngeal carcinoma (hematoxylin and eosin, $\times 200$). (b) Histology slide showing type II nonkeratinizing squamous cell nasopharyngeal carcinoma (hematoxylin and eosin, $\times 200$). (c) Histology slide showing type III undifferentiated nasopharyngeal carcinoma (hematoxylin and eosin, $\times 200$). (Used with permission from Lund VJ, Howard DJ, Wei WJ. *Tumors of the Nose, Sinuses, and Nasopharynx*. Stuttgart/New York; Thieme; 2014:539.)

EBV DNA titers seem to be an important index for prognostication. EBV DNA titers correlate with stage, treatment response, relapse, and survival.

Adoptive immunotherapy with EBV-specific CTL (cytotoxic T cell lymphocytes) awaits further exploration.

Surgical

The role of surgery in NPC is largely confined to the treatment of residual or recurrent disease either in the nasopharynx or in the neck, though there is literature from China discussing the use of surgery as a primary-modality treatment. Neck dissection for postradiation residual or recurrent nodal disease is the most common indication for surgery.

◆ Outcome and Follow-Up

Overall survival with the use of conventional radiotherapy alone in the treatment of NPC is in the range of 50 to 76%. Patients with stage I and II disease have a high rate of cure with radiotherapy alone. Seventy percent of patients with NPC present with locally advanced stage III or IV disease. For these patients, radiotherapy delivered in combination with chemotherapy has become the standard of care. The prognosis for those with distant metastatic spread remains poor.

◆ Staging of Nasopharyngeal Cancer

Primary Tumor (T)

- TX: Primary tumor cannot be assessed
- T0: No tumor identified, but EBV-positive cervical node(s) involvement
- Tis: Carcinoma in situ
- T1: Tumor confined to nasopharynx, or extension to oropharynx and/or nasal cavity without parapharyngeal involvement
- T2: Tumor with extension to parapharyngeal space, and/or adjacent soft tissue involvement (medial pterygoid, lateral pterygoid, prevertebral muscles)

- T3: Tumor with infiltration of bony structures at skull base, cervical vertebra, pterygoid structures, and/or paranasal sinuses
- T4: Tumor with intracranial extension, involvement of cranial nerves, hypopharynx, orbit, parotid gland, and/or soft tissue infiltration beyond the lateral surface of the pterygoid muscle

Regional Lymph Node (N)

- NX: Regional lymph nodes cannot be assessed
- N0: No regional lymph node metastasis
- N1: Unilateral metastasis in cervical lymph node(s) and/or unilateral or bilateral metastasis in retropharyngeal lymph node(s), ≤ 6 cm in greatest dimension, above the caudal border of cricoid cartilage
- N2: Bilateral metastasis in cervical lymph node(s), ≤ 6 cm in greatest dimension, above the caudal border of cricoid cartilage
- N3: Unilateral or bilateral metastasis in cervical lymph node(s), > 6 cm in greatest dimension, and/or extension below the caudal border of cricoid cartilage

Distant Metastasis (M)

- M0: No distant metastasis
- M1: Distant metastasis

American Joint Committee on Cancer Stage Groupings of Nasopharynx Cancers (8th Edition)

- Stage 0 disease: Carcinoma in situ (TisN0M0)
- Stage I disease: Includes only T1 N0 M0 tumors
- Stage II disease: Includes T0–N1M0 and T2 N0–N1 M0 tumors
- Stage III disease: Includes T3 N0 M0 and T0–3 disease that is N2 M0
- Stage IVA disease: Includes T4 disease with N0–N2 M0 disease and T0–T4 disease that is N3 M0
- Stage IVB disease: Any disease with distant metastases (M1) (Table 6.10)

Table 6.10 Cancer Stage Groupings for Nasopharynx Cancer

	N0	N1	N2	N3
T0	N/A	Stage II	Stage III	Stage IVA
T1	Stage I	Stage II	Stage III	Stage IVA
T2	Stage II	Stage II	Stage III	Stage IVA
T3	Stage III	Stage III	Stage III	Stage IVA
T4	Stage IVA	Stage IVA	Stage IVA	Stage IVA
M1	Stage IVB	Stage IVB	Stage IVB	Stage IVB

6.3.5 Oral Cavity Cancer

◆ Key Features

- A nonhealing or bleeding sore in the mouth or on the lip is the most common presentation.
- A persistent white or red patch on oral mucosa needs to be investigated.
- History of tobacco smoking or chewing occurs in most cases.
- Lip cancer may be caused by sun exposure.

The oral cavity extends from the vermilion borders of the lips anteriorly to the junction of the hard and soft palates superiorly and to the line of circumvallate papillae posteriorly. Oral cancer should be identified early, and screening is useful. It is frequently preceded by an identifiable premalignant lesion. The progression from dysplasia may occur over a period of years.

◆ Epidemiology

Thirty thousand people are diagnosed yearly with oral cancer in the United States, and it causes > 8,000 deaths. It constitutes < 5% of US cancers, but in India, the incidence is far greater. For all stages combined, the 5-year relative survival rate is 59%, and the 10-year survival rate is 44%. It typically occurs in those over the age of 45 and occurs in men twice as often as in women. The number of new cases of this disease has been decreasing during the past 20 years. Tobacco smoking and alcohol are the primary risk factors. Smokeless tobacco in the Western world and paan (betel leaf with areca nut) in Asia are also risk factors for oral cancer. Other suspected risk factors include viral infection (human papillomavirus [HPV]), a diet low in fruits and vegetables, vitamin A deficiency, genetic susceptibility, and immunosuppressive drugs or immunocompromised conditions. Recently there has been a growing number of young patients with oral cancers, particularly involving the tongue.

◆ Clinical

Signs

Signs may include uncomfortable or poorly fitting dentures, loosening of the teeth, changes in articulation, a mass in the neck, weight loss, and persistent halitosis.

Symptoms

Symptoms depend on site and stage of the primary tumor and its effect on function of that area. They include a nonhealing white or red (leukoplakia, erythroplakia) patch or sore in the mouth (most common symptom), persistent pain in the mouth, and a thickening in the cheek or the floor of the mouth. More advanced disease may cause a sore throat, difficulty chewing,

dysphagia, trismus or tongue tethering, numbness of the tongue or mouth, and pain around the teeth or jaw.

Differential Diagnosis

- Oral and pharyngeal infections such as pharyngitis or stomatitis
- Chancre
- Benign oral or odontogenic lesions
- Denture sores
- Aphthous ulcers or herpetic sores
- Lesion due to cheek biting
- Oral manifestations of systemic diseases
- Necrotizing sialometaplasia

◆ Evaluation

History

Evaluation begins with a detailed history inquiring about tobacco and alcohol usage, oral pain, referred otalgia, dysphagia, articulation changes, and weight loss.

Physical Exam

A physical exam should include a complete head and neck exam. Specific attention should be directed at the site of the lesion. The lesion size should be noted, as should its infiltration and spread to adjacent oral cavity or oropharyngeal subsites such as the floor of the mouth, alveolus, and tongue base. A bimanual examination of the lesion, the surrounding floor of the mouth, and the submandibular triangle should be performed.

Careful palpation of the neck may reveal adenopathy. The main routes of lymph node drainage from the oral cavity are into the first-echelon nodes (i.e., buccinator, jugulodigastric, submandibular, and submental). Tumor sites close to the midline commonly drain bilaterally.

Imaging

Contrast-enhanced computed tomography (CT) of the head and neck is a necessary component of initial evaluation. Tumor size and spread may be evaluated as well as discrete nodal disease, bony destruction, and vascular involvement. Magnetic resonance imaging (MRI) may be helpful in the evaluation of oral cancer because it provides a higher contrast between normal tissue and tumor on T2-weighted images, has no beam artifact from dental material, and provides multiplanar imaging.

The combination of positron emission tomography (PET) and CT is a useful diagnostic and staging modality in the evaluation of the patient with head and neck cancer.

A chest CT may be used to rule out pulmonary metastasis. Periapical dental radiographs provide fine detail and may show minimal invasion. Panoramic dental radiographs may show gross bony destruction.

Labs

Routine preoperative laboratory studies are employed.

Other Tests

Patients with suspected oral cancer must undergo a biopsy for pathologic diagnosis. The first test for the evaluation of a neck mass presenting with an oral cavity lesion is fine-needle aspiration biopsy (FNAB).

The routine use of panendoscopy, which includes bronchoscopy, esophagoscopy, and laryngoscopy, is recommended. It allows for the complete evaluation of the upper aerodigestive tract and helps rule out the presence of a synchronous tumor. The mucous membranes of the upper aerodigestive tract are carefully evaluated, and biopsy samples of any abnormal-looking areas are taken. This is particularly important in the patient who smokes.

Toluidine blue staining and photodynamic agents such as 5-aminolevulinic acid (ALA) may be used to enhance detection of oral lesions. Oral brush biopsy (OralCDx; CDx Diagnostics, Suffern, NY) can be used in the screening of oral precancer and cancer.

A dental evaluation should be performed, with attention to dental hygiene, dentition status, and integrity of the mandible.

Pathology

Ninety percent of oral cancers are squamous cell carcinomas (SCCs), and they may be preceded by various precancerous lesions. Non-SCC oral cancers may also include minor salivary gland tumors. Other rarer cancers may be of odontogenic apparatus origin, lymphomas, soft tissue sarcomas, or melanomas.

There is no significant correlation between degree of squamous differentiation (see **Table 6.5**) and the biologic behavior of oral cancer. Vascular and perineural invasion thickness and depth of invasion are all negative prognostic factors.

Verrucous carcinoma (VC) is a locally aggressive, clinically exophytic, low-grade, well-differentiated SCC with minimal metastatic potential. It is also known as Ackerman's tumor. An HPV infection is thought to facilitate or cause verrucous carcinoma. Associations with verrucous carcinoma have been found in patients who chew tobacco and betel nut. Fully developed lesions are white cauliflowerlike papillomas with a pebbly surface that may extend and coalesce over large areas of the oral mucosa. Overall, patients with verrucous carcinoma have a favorable prognosis; the course of verrucous carcinoma lesions is characterized by slow, continuous local growth.

Proliferative verrucous leukoplakia (PVL) is a particularly aggressive form of oral leukoplakia that commences with a hyperkeratosis, spreads to become multifocal and verruciform in appearance, and later becomes malignant. It is significant because it has a high recurrence rate and the potential to develop into verrucous carcinoma or SCC in 60 to 70% of the affected patients. PVL is more commonly found in elderly females and is associated with tobacco use or alcohol abuse in 30 to 50% of patients. The etiology of PVL is unknown. An association with HPV infection, particularly strains 16 and 18, has been implicated in some cases. The most common

locations are the gingiva or alveolar ridge, the tongue, and the buccal mucosa. The gingiva is the most likely site for the malignant transformation of PVL. PVL often begins as a focal lesion spreading laterally over time and can be multifocal. Early in its course, it is a flat hyperkeratotic lesion that becomes progressively verrucous and histologically often exhibits varying degrees of epithelial dysplasia.

◆ Treatment Options

Surgical resection and radiotherapy are the current treatments of choice. Surgery is considered the primary and preferred modality of treating cancers of the oral cavity.

Lip Cancer

Most lip SCCs present on the lower lip (88–95%), 2 to 7% present on the upper lip, and 1% on the oral commissure. Basal cell carcinoma is more common on the upper lip.

Treatment of Lip Cancer

- For T1 and T2 lesions, radiotherapy and surgery produce similar cure rates; the method of treatment is determined by functional and cosmetic factors.
- Advanced lesions of the lip generally require a combination of surgery and radiotherapy.
- Patients with upper lip and oral commissure SCC have a worse overall prognosis.
- The 5-year survival for stage I and II lesions is 90%.

Oral Tongue Cancer

Seventy-five percent of tongue cancers occur on the posterior lateral aspect, 20% on the anterior lateral aspect, and 3 to 5% on the lingual dorsum. At the time of diagnosis, 75% oral tongue cancers are T2 or smaller. Forty percent of patients with oral tongue cancer demonstrate clinical evidence of neck metastasis at presentation.

Treatment of Oral Tongue Lesions

- Early tongue cancer: Wide local excision is often used for T1 lesions that can be resected transorally.
- For larger T1 and T2 lesions, either surgery or radiotherapy is an acceptable treatment.
- Deeply infiltrative lesions (> 4 mm depth) can be treated with surgery with postoperative radiotherapy and a selective neck dissection.
- Selected patients with T4 tongue cancer can be treated with combined surgery (i.e., total glossectomy, sometimes requiring laryngectomy due to the high risk of postoperative aspiration) and postoperative radiotherapy.
- For T1 and T2 lesions, 20 to 30% of patients harbor metastatic disease in cervical lymph nodes. Thus, therapy (surgery or radiotherapy) aimed at the neck should be considered as part of definitive treatment.

- The 5-year survival is 75% for stage I and II oral tongue cancers and < 40% for stage III and IV oral tongue cancers.

Buccal Mucosa Cancer

Carcinomas of the buccal mucosa represent 5 to 10% of oral cancers. The most common area is in the region of the third mandibular molar. Lesions < 1 cm in diameter can be treated by surgery alone if the oral commissure is not involved. If involved, radiotherapy should be considered. Premalignant conditions include submucosal fibrosis and lichen planus. The latter has a reported transformation rate of 0.5 to 3%, whereas the former has a malignant transformation rate of 0.5%.

Treatment of Buccal Mucosa Cancers

- Lesions smaller than 1 cm in diameter can be managed by surgery alone if the commissure is not involved. If the commissure is involved, radiotherapy (including brachytherapy) should be considered.
- Advanced lesions of the buccal mucosa can be treated with surgical resection alone, radiotherapy alone, or surgical resection plus postoperative radiation.
- The 5-year survival for buccal mucosa cancer is 75% for stage I, 65% for stage II, 30 to 65% for stage III, and 20 to 50% for stage IV buccal cancer.

Floor of Mouth Cancer

Cancers of the floor of the mouth represent 28 to 35% of oral cancers. Thirty-five percent of patients with floor of mouth cancer present with T3 or T4 disease. The most common presentation of cancer of the floor of the mouth is a painless inflamed superficial ulcer with poorly defined margins. Preexistent or coincident leukoplakia can be observed in adjacent tissues in ~ 20% of cases.

Treatment of Floor of Mouth Cancer

Note that a cancer involving the gingiva adjacent to recent dental extraction is at high risk for bony extension via the tooth socket.

- For T1 lesions, either transoral surgery or radiotherapy is an acceptable treatment.
- For small T2 lesions (≤ 3 cm), surgery may be used if the lesion is attached to the periosteum, whereas radiotherapy may be used if the lesion encroaches on the tongue.
- For large T2 lesions (> 3 cm), surgery and radiotherapy are alternative methods of treatment, the choice of which depends primarily on the expected extent of disability from surgery.
- External-beam radiotherapy with or without interstitial radiotherapy should be considered postoperatively for larger lesions.
- For more advanced lesions, surgery should incorporate rim resection plus neck dissection or partial mandibulectomy with neck dissection as appropriate.

- The 5-year survival for floor of mouth cancer is 90% for stage I, 80% for stage II, 65% for stage III, and 30% for stage IV.

Retromolar Trigone Cancer

Retromolar trigone cancers account for ~ 10% of all oral cancers. These cancers typically present with advanced disease, and 50% of patients have regional metastasis at the time of diagnosis.

Treatment of Retromolar Trigone Cancer

For small lesions without detectable bone invasion, limited resection of the mandible may be performed. Radiotherapy may be used initially, with surgery reserved for radiation failure. Selective neck treatment should be performed—for advanced stages, multimodality therapy with surgery and postoperative radiation is most often used.

Hard Palate Cancer

Cancer of the hard palate accounts for 5% of all oral cavity malignancies. Ten to 25% of patients with head and neck SCC of the hard palate present with regional metastasis. (Only 53% of hard palate cancers are SCC; minor salivary gland malignancies make up the rest.)

Treatment of Cancer of the Hard Palate

For both early and advanced disease, surgery (inferior maxillectomy with surgical obturator) is used for primary therapy. Radiotherapy has a role depending on factors such as close or positive surgical margins, evidence of perineural involvement, or the presence of lymph node metastases. The prosthodontist is important in the care of these patients for oral rehabilitation. The 5-year survival for hard palate cancer ranges from 40 to 60%.

Advanced Oral Cavity Cancer

Clinical trials for advanced oral tumors evaluating the use of chemotherapy preoperatively, before radiotherapy, as adjuvant therapy after surgery, or as part of combined-modality therapy are appropriate.

◆ Outcome and Follow-Up

The National Comprehensive Cancer Network (NCCN) guidelines suggest routine follow up after treatment. This includes follow up every 1 to 2 months the first year, every 2 to 6 months the second year, every 4 to 8 months the third through fifth years and annual follow up afterwards. Patients with regional neck disease prior to treatment should undergo CT scan or integrated ^{18}F fludeoxyglucose (FDG)-PET/CT 12 weeks after the completion of radiotherapy to assess for residual disease that may necessitate postradiotherapy neck dissection. The risk of a second primary carcinoma is highest in those who continue to smoke, and patients should be strongly urged to quit.

See **Fig. 6.12**.

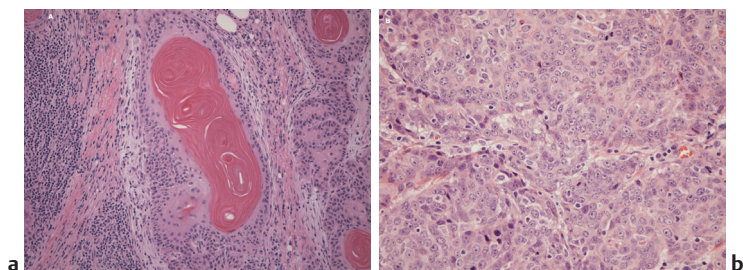


Fig. 6.12 Squamous cell carcinoma. **(a)** Well-differentiated squamous cell carcinomas are often extensively keratinized (a keratin pearl is seen in the center of a nest), with mild cytologic atypia and mitotic activity limited to the parabasal zones. **(b)** In contrast, poorly differentiated squamous cell carcinomas grow as cords or sheets of highly pleomorphic cells with frequent mitoses. There is little evidence of maturation within the tumor nests. (Used with permission from Har-El G, Day T, Nathan C-A, Nguyen SA. *A Multidisciplinary Approach to Head and Neck Neoplasms*. New York: Thieme;2013. *Otorhinolaryngology—Head and Neck Surgery Series*.)

◆ Staging of Oral Cavity Cancer

Primary Tumor (T) (now includes Depth of Invasion [DOI])

TX: Primary tumor cannot be assessed

Tis: Carcinoma in situ

T1: Tumor ≤ 2 cm in greatest dimension, and ≤ 5 mm DOI

T2: Tumor ≤ 2 cm and DOI > 5 mm and ≤ 10 mm or tumor > 2 cm and ≤ 4 cm, and ≤ 10 mm DOI

T3: Tumor > 4 cm in greatest dimension or any tumor > 10 mm DOI

T4a: Moderately advanced local disease

Lip: Tumor invades through the cortical bone, the inferior alveolar nerve, the floor of the mouth, or the skin of the face (i.e., the chin or nose).

Oral cavity: Tumor invades the adjacent structures, such as the cortical bone [mandible, maxilla], the maxillary sinus, or the skin of the face.

T4b: Very advanced local disease. Tumor invades the masticator space, pterygoid plates, or skull base and/or encases the internal carotid artery.

Regional Lymph Nodes (N)*

NX: Regional lymph nodes cannot be assessed

N0: No regional lymph node metastasis

N1: Metastasis to a single ipsilateral lymph node measuring ≤ 3 cm in greatest diameter and ENE negative

N2: Further divided into three categories:

N2a: Single ipsilateral lymph node > 3 and ≤ 6 cm and ENE negative (pathologic staging also includes a single ipsilateral or contralateral lymph node ≤ 3 cm and ENE positive)

- N2b: Multiple ipsilateral lymph nodes ≤ 6 cm and ENE negative
- N2c: Bilateral or contralateral lymph nodes ≤ 6 cm in greatest dimension and ENE negative

N3: Now further divided into two categories:

- N3a: Metastasis to a lymph node > 6 cm, ENE negative
- N3b: Metastasis to a single lymph node that is ENE positive (pathologic staging includes single lymph nodes that are ENE positive >3cm), or metastasis to multiple lymph nodes with any ENE positive

A designation of “U” or “L” can be attached to the N category to indicate nodes above (“U”) or below (“L”) the lower border of the cricoid. Similarly, clinical and pathologic ENE should be recorded as ENE (–) or ENE (+). Clinical and pathologic staging are similar with the differences noted in parenthesis.

*Superior mediastinal lymph nodes are considered regional lymph nodes (level VII). Midline nodes are considered ipsilateral nodes.

Distant Metastasis

- M0: No distant metastasis
- M1: Distant metastasis

American Joint Committee on Cancer Stage Groupings for Oral Cavity Cancers (8th edition)

- Stage 0 disease: Carcinoma in situ (TisN0M0)
- Stage I disease: Includes only T1 N0 M0 tumors
- Stage II disease: Includes only T2 N0 M0 tumors
- Stage III disease: Includes T3 N0 M0 and T1 -3 disease that is N1 M0
- Stage IVA disease: Includes T4a disease with N0-N2 M0 disease and T1-T3 disease that is N2 M0
- Stage IVB disease: Includes T4b disease with any nodal disease or T1-T4a disease with N3 disease without distant metastases (M0)
- Stage IVC disease: Any disease with distant metastases (M1) (Table 6.11)

Table 6.11 Cancer stage groupings

	N0	N1	N2	N3
T1	Stage I	Stage III	Stage IVA	Stage IVB
T2	Stage II	Stage III	Stage IVA	Stage IVB
T3	Stage III	Stage III	Stage IVA	Stage IVB
T4a	Stage IVA	Stage IVA	Stage IVA	Stage IVB
T4b	Stage IVB	Stage IVB	Stage IVB	Stage IVB
M1	Stage IVC	Stage IVC	Stage IVC	Stage IVC